Mend-Amar Badral

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Driven by a problem solving mindset and fueled by curiosity, I always aspire to develop solutions for the real-world, impactful problems.

Education

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering Budapest, Hungary
Supervisor: Márton Vaitkus
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics Ulaanbaatar, Mongolia
GPA: 3.2/4.0 **(rank 3/34)**
Supervisor: Galtbayar Artbazar
Thesis: *Evaluating land degradation using image processing methods*

Experience

Center of Mathematics Application, NUM Oct, 2023 – June, 2025
Undergraduate Research Intern

- **Technical Documentation:** Authored and maintained comprehensive lab documentation and user guides, enabling seamless onboarding for new students and establishing clear protocols for research operations. Served as Lab Lead for a team of 6-8 students; collaborated to distill and present complex mathematical concepts during weekly seminars.
- **Data Management:** Architected a unified, well-documented data folder structure for research backups, directly improving data retrieval efficiency and creating written guidelines for future data integration.
- **Programming:** Engineered a custom a Python program for processing drone imagery which enabled easy method to obtain necessary vegetation index from drone imagery data.

Projects

aiSim Data Integration for 3D Gaussian Splatting

- Implemented a JSON parser in Python to analyze and convert complex sensor measurements from aiMotive's aiSim simulation software into a 3D scene representation (nerfstudio) format.

Short Animation: What is a Derivative?

- Created a short-form educational video explaining the mathematical concept of derivatives utilizing Python Manim library. Achieved 100 views on the YouTube platform.

Skills

Languages:	English (C1 IELTS 8.0), Mongolian (native), Hungarian (A2)
Programming:	Python, Bash, Julia, C/C++, R, MATLAB
Tools:	Git, GitHub, Linux, Docker, QGIS, Manim, LaTeX